

Erdem Aslan

Computer Engineering Student · AI / ML · Full-Stack Development

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WORK EXPERIENCE

- Full-Stack Developer Intern** | [Proceedit Trading, Barcelona, Spain \(Remote\)](#) Oct 2025 – May 2026
- Developed backend API components using Python (Flask) and PostgreSQL; implemented REST endpoints and database integrations, improving system data flow and reliability.
 - Managed pod and service operations on AWS and Kubernetes environments; accelerated development, debugging, and backend testing processes.
 - Contributed to AI-driven algorithmic modeling on financial time series data; supported performance analysis of CatBoost and LSTM-based models in production.
 - Managed sprint tasks via JIRA and coordinated with an international team through Google Meet; actively participated in Agile ceremonies to improve delivery speed.
 - Prototyped cross-platform mobile interfaces using FlutterFlow and Figma; developed design solutions that enhanced user experience.
- iOS Application Developer** | [Lideatech](#) Jun 2025 – Jul 2025
- Built FinFlow, a personal finance app from scratch using Swift/SwiftUI and Firebase; implemented secure real-time data flow with Firestore integration and user authentication.
 - Developed income/expense tracking, budget analysis, and data visualization features using modern iOS practices; improved UX with interactive dashboards.
 - Adopted MVVM architecture to establish a modular and maintainable codebase; ensured long-term scalability and sustainability of the application.
- Game Operator & Cook** | [Morey's Piers Amusement Park, USA \(Work & Travel\)](#) Jun 2024 – Sep 2024
- Gained international work experience in the USA through the Work & Travel programme; adapted quickly to a fast-changing environment, contributing to operational continuity.
 - Collaborated with teammates from diverse cultural backgrounds; significantly improved English communication skills and cross-cultural awareness.

AI / LLM PROJECTS

- LLM Distillation & Quantization** | [Bachelor's Thesis](#) · [arXiv:2601.16219](#) 2025
- Developed a domain-specific LLM optimization pipeline using knowledge distillation and quantization techniques.
 - Reduced model size and computational cost while maintaining performance for resource-constrained deployment.
 - Implemented fine-tuning strategies on modern LLM architectures; research published on arXiv.
- Historai — Multi-Agent Historical Simulation** | [FastAPI](#) + [Next.js](#) + [Groq](#) + [Supabase](#) 2026
- Built a counterfactual simulation engine powered by LLMs; multi-agent architecture with Groq (LLaMA 3.3 70B) and Gemini fallback.
 - Implementing RAG pipeline with vector embeddings and Supabase pgvector for context-aware historical retrieval.
 - Generates historical actor personas, narrative timelines, and interactive scenario analysis.
- GAN-based Data Augmentation** | [Deep Learning / Medical Imaging](#) 2024
- Built a DCGAN pipeline (PyTorch) to generate synthetic CT images; improved classification performance via class balancing.

EDUCATION

- B.Sc. Computer Engineering** | [Düzce University](#) 2022 – Present
- GPA: 3.14 / 4.00 · Completed 4-year curriculum in 3.5 years · 1 year English prep (2021–2022)
- Erasmus+ Exchange** | [Gdańsk University of Technology](#) Feb 2026 – Jun 2026
- Fully funded Erasmus+ mobility grant

TECHNICAL SKILLS

AI / LLM: LLM distillation, quantization, fine-tuning (LoRA/QLoRA), RAG pipeline, vector embeddings, prompt engineering
Frameworks: PyTorch, LangChain, LangGraph, CrewAI, Hugging Face, Groq API, OpenAI API, Gemini API
Vector DB: Supabase pgvector, Chroma (basic), embedding pipelines
Backend: Python, Flask, FastAPI, REST APIs, PostgreSQL, Firebase
Cloud / DevOps: AWS, Kubernetes, Docker, Prometheus, Grafana, OpenTelemetry, Linux, Git
Languages: Python, SQL, Swift, Java, C#, Bash

CERTIFICATIONS & TRAINING

- Application Monitoring with OpenTelemetry** | [Mustafa Akgül Open Source Software Camp](#) Feb 2026
- 4-day hands-on training: Prometheus, Grafana, OpenTelemetry SDK/Collector, distributed tracing in Kubernetes environments.
- Intro to Blockchain & Ethereum Smart Contracts** | [Mustafa Akgül Open Source Software Camp](#) Aug 2025
- 9-day intensive program (74 hours) covering blockchain fundamentals and Ethereum smart contract development.

LANGUAGES

English: Full Professional Proficiency (C1)
Turkish: Native